



DIGITAL LOGIC

Logic Function Simplification

Dr. Fei Chao
Associate Professor
Cognitive Science Department
School of Information Science and Technology
Xiamen University
fchao@xmu.edu.cn

2.4 Simplification/Optimization of logic functions

The complexity of a logic circuit realizing a logical function is directly related to the complexity of the logic expression describing the function.

In order to reduce system cost, reduce complexity and improve reliability, logic functions must be simplified.

Because the “AND-OR“ and “OR-AND” expressions can be easily converted to any other required form. Therefore, we can start from these two common forms, to solve the problem of function simplification, and focus on simplification of “AND-OR” expression.

There are 3 common methods to simplify logical functions. That is, algebra simplification, Karnaugh map simplification and Quine-McCluskey tabular simplification.

2.4.1 Algebra simplification method

Algebra simplification is the way to simplify logical functions by using properties, theorems, and rules of logic algebra.

I. Simplification of “AND-OR” forms

Simplest “AND-OR” expression must meet two criteria:

- 1. The number of “AND” terms in the expression is the least;**
- 2. Under the premise of satisfying the above condition, the number of variables in each AND term is the least.**

Satisfying the above two criteria can make the number of required gates and the number of gate inputs in the corresponding logic circuits be the minimum, thereby making the circuit the most economical.

Common methods:

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1. Combining term method

Use Theorem 7: $\overline{A}B + AB = A$, Combine the two "AND" terms into an "AND" term, merge and eliminate one variable. E.g.

$$\overline{A}BC + \overline{A}B\overline{C} = \overline{A}B$$

2. Absorption method

$$\overline{A}B(C + \overline{C}) = \overline{A}B$$

Use Theorem 3 $\overline{A} + \overline{A}B = \overline{A}$, absorb redundant terms. E.g.,

$$\overline{A}B + \overline{A}B\overline{C} = \overline{A}B$$

//

$$\overline{E} + \overline{E}C = \overline{E}$$

$$AB + \bar{A}\bar{C} + B\bar{C} + \bar{B}\bar{C}$$

3. Elimination method

$$\begin{aligned} & \bar{C}(B + \bar{B}) = AB + \bar{A}\bar{C} + \bar{C} \\ & = AB + \bar{C} \end{aligned}$$

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Use Theorem 4 $A + \bar{A}B = A + B$ to eliminate redundant variables.

E.g.,

$$\underline{AB} + \underline{\bar{A}\bar{C}} + \underline{\bar{B}\bar{C}} = AB + (\bar{A} + \bar{B})\bar{C} = \underline{AB} + \underline{\bar{A}\bar{B}\bar{C}} = \underline{AB + \bar{C}}$$

4. Term coupling method

$$\bar{C} + \bar{C} = \bar{C} \quad \bar{C} + \bar{C} = \bar{C}$$

Use Properties 4 and 5 $A \cdot 1 = A$, $A + \bar{A} = 1$, firstly, some of the "AND" terms are properly selected from the function, coupled with a suitable variable missing from them, and then simplified by using combining, absorption and elimination methods. E.g,

$$\begin{aligned} & \underline{\bar{A}\bar{B}} + \underline{\bar{B}\bar{C}} + \underline{\bar{B}\bar{C}} + \underline{\bar{A}\bar{B}} \\ & = \underline{\bar{A}\bar{B}} + \underline{\bar{B}\bar{C}} + (A + \bar{A})\bar{B}\bar{C} + \underline{\bar{A}\bar{B}} \cdot (C + \bar{C}) \\ & = \underline{\bar{A}\bar{B}} + \underline{\bar{B}\bar{C}} + \underline{\bar{A}\bar{B}\bar{C}} + \underline{\bar{A}\bar{B}\bar{C}} + \underline{\bar{A}\bar{B}\bar{C}} + \underline{\bar{A}\bar{B}\bar{C}} \\ & = \underline{\bar{A}\bar{B}} + \underline{\bar{B}\bar{C}} + \underline{\bar{A}\bar{C}} \end{aligned}$$

$$\bar{A}\bar{C}(\bar{B} + B) = \bar{A}\bar{C}$$

The logic functions encountered in practical applications are often more complex, and the properties, theorems, and rules learned should be used flexibly when simplification, and various methods should be comprehensively used.

E.g. Simplify $F = BC + D + \bar{D} \cdot (\bar{B} + \bar{C}) \cdot (AD + B\bar{C})$

A:

$$BC + D + (\bar{B} + \bar{C})(AD\bar{D} + B\bar{C}\bar{D})$$

$$= BC + D + \bar{B}\bar{C} \cdot B\bar{C}\bar{D}$$

$$= \bar{B} + D + \bar{B}\bar{C}\bar{D}$$

$$= \bar{B} + D$$

$$F = BC + D + \bar{D} \cdot (\bar{B} + \bar{C}) \cdot (AD + B\bar{C})$$

$$= BC + D + (\bar{B} + \bar{C}) \cdot (AD + B\bar{C})$$

$$= BC + D + \overline{BC} \cdot (AD + B\bar{C})$$

$$= BC + D + \overline{AD + B\bar{C}}$$

$$= \underline{B + D}$$

$$A + \bar{A}B = A + B$$

$$\bar{B}\bar{C} + \bar{B}\bar{C} = 1$$

E.g. Simplify

$$F = \overline{\overline{A(B + \overline{C})}(A + \overline{B} + C)\overline{\overline{ABC}}}$$

A: $F = \overline{\overline{A(B + \overline{C})} \cdot (A + \overline{B} + C) \overline{\overline{ABC}}}$

$$= \overline{(A + \overline{B} + \overline{C})(A + \overline{B} + C)(A + B + C)}$$

$$\Rightarrow (A + \overline{B}C) \cdot (A + C)$$

$$= A + \overline{B}CC$$

$$= A + \overline{B}C$$

$$= A\overline{B}C + AB C$$

$$= AC(\overline{B} + B)$$

$$= AC$$

$$= (A + C)$$

$$A + \overline{B}C = (A + B)(A + C)$$

II. Simplification of “OR-AND” forms

Simplest “OR-AND” expression must meet two criteria:

- 1. The number of “OR” terms in the expression is the least;**
- 2. Under the premise of satisfying the above conditions, the number of variables in each OR term is the least.**

The algebra simplification of “OR-AND” expressions can be used directly in the form of “OR-AND” in properties and theorems, and the various methods proposed in the previous method of “AND-OR” expression simplification methods.



E.g.**Simplify**

$$F = (A + B) \cdot (\bar{B} + C) \cdot (A + C + \bar{D}) \cdot (A + C)$$

$$\text{A: } F = (A + B) \cdot (\bar{B} + C) \cdot (A + C + \bar{D}) \cdot (A + C)$$

$$= (A + B) \cdot (\bar{B} + C) \cdot (A + C)$$

$$= (A + B)(\bar{B} + C)$$

$$(A + B + \bar{B}C) + AC$$

$$AB + \bar{A}C + BC = AB + \bar{A}C$$

In addition, we can use the double dual method, which is:

Step 1: Dual the F with “OR-AND” forms, to obtain F’ with “AND-OR” forms;

Step 2: Simplify F’ to have the simplest “AND-OR” forms;

Step 3: Dual F’ again, so as to obtain the simplest F with “OR-AND” forms.

E.g. Simplify $F = (A + \bar{B}) \cdot (\bar{A} + B) \cdot (B + C) \cdot (\bar{A} + C)$

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A: Step 1: Dual the F to have F':

$$F' = \bar{A}\bar{B} + \bar{A}B + BC + \bar{A}C$$

Step 2: Simplify F' :

$$\begin{aligned} F' &= A\bar{B} + \bar{A}B + BC + \bar{A}C \\ &= A\bar{B} + \bar{A}B + (B + \bar{A})C \\ &\stackrel{E}{=} A\bar{B} + \bar{A}B + \bar{A}BC \\ &= A\bar{B} + \bar{A}B + C \end{aligned}$$

$\bar{E} + \bar{E}C$

Step 3: Dual F' again to obtain the simplest F with "OR-AND" forms.

$$F = (A + \bar{B}) \cdot (\bar{A} + B) \cdot C$$

The advantage of algebra simplification is that it is not constrained by the number of variables. If we can be very familiar with the properties, theorems and rules are, such simplification method is more convenient.

The disadvantages are: there are no certain laws and steps, and the skills are very strongly required, and in many cases it is difficult to determine whether the simplified results are the simplest.

2.4.2 Karnaugh Maps (K-map)

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The Karnaugh map simplification method has the advantages of being simple, intuitive and easy to learn, and has been widely used in logic design.

I. Components of K-Map

A K-map is a collection of squares

Each square represents a Minterm

The collection of squares is a graphical representation of a Boolean function

A K-map with n variables has 2^n

Adjacent squares differ in the value of one variable

Alternative algebraic expressions for the same function are derived by recognizing patterns of squares

The K-map can be viewed as

A reorganized version of the truth table

Two Variable Maps

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- A 2-variable Karnaugh Map:

- Note that minterm m_0 and minterm m_1 are “adjacent” and differ in the value of the variable y

- Similarly, minterm m_0 and minterm m_2 differ in the x variable.

- Also, m_1 and m_3 differ in the x variable as well.

- Finally, m_2 and m_3 differ in the value of the variable y

	$y = 0$ 0	$y = 1$ 1
$x = 0$ 0	$m_0 = \overline{x}y$	$m_1 = x\overline{y}$
$x = 1$ 1	$m_2 = x\overline{y}$	$m_3 = xy$

K-Map and Truth Tables

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- The K-Map is just a different form of the truth table.
- Example – Two variable function:
 - ▣ We choose a, b, c and d from the set $\{0,1\}$ to implement a particular function, $F(x,y)$.

Function Table

Input Values (x,y)	Function Value $F(x,y)$
0 0	a
0 1	b
1 0	c
1 1	d

K-Map

	$y = 0$	$y = 1$
$x = 0$	a m_0	b .
$x = 1$	c m_1	d

2-variable, 3-variable, 4-variable Karnaugh map:

B \ A	0	1
0	m_0	m_2
1	m_1	m_3

(a)

C \ AB	00	01	11	10
0	m_0	m_2	m_6	m_4
1	m_1	m_3	m_7	m_5

(b)

CD \ AB	00	01	11	10
00	m_0	m_4	m_{12}	m_8
01	m_1	m_5	m_{13}	m_9
11	m_3	m_7	m_{15}	m_{11}
10	m_2	m_6	m_{14}	m_{10}

(c)

By analyzing these K-maps, **we can intuitively and conveniently find each minterm's n adjacent minterms from the Karnaugh map of n variables.**

E.g., a 4-variable K-map, the m_5 has 4 adjacent minterms, which are m_1 , m_4 , m_7 , m_{13} .

The m_2 has 4 adjacent minterms, including geometrically adjacent minterms: m_3 and m_6 ; also, including m_0 (two ends of the same column) and m_{10} (two ends of the same row). These situation are also belonged to the adjacent positions.

		AB			
		00	01	11	10
CD	00	m_0	m_4	m_{12}	m_8
	01	m_1	m_5	m_{13}	m_9
	11	m_3	m_7	m_{15}	m_{11}
	10	m_2	m_6	m_{14}	m_{10}

Furthermore, in the following 5-variable K-map, it is easy to find the geometrically adjacent minterms of m_3 ; i.e., m_1 , m_2 , m_7 and m_{11} , it also has m_{19} as an adjacent minterm.

5-variable K-map

DE \ ABC		000 001 011 010				100 101 111 110			
		000	001	011	010	100	101	111	110
00		0	4	12	8	16	20	28	24
01		1	5	13	9	17	21	29	25
11		3	7	15	11	19	23	31	27
10		2	6	14	10	18	22	30	26

(d)

II. Properties of K map

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Property: we can intuitively find out the adjacent minterms to combine using the minimization theorem.

$\overline{A}\overline{B} + AB = A$

	AB	00	01	11	10
CD	00				
	01		m_5	m_{13}	
	11		m_7	m_{15}	
	10				

$$\frac{\overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D}{\overline{A}BD + ABD}$$

$B D$

The basic principle of simplifying the logical function in K map is to combine the adjacent small squares of adjacent minterms on the K map to merge together to achieve the purpose of replacing a number of minterms with a simple "AND" term.

It is usually called the "K Rectangle" that surrounds those "minimum" that can be replaced by a simple "AND" item.

III. K-Map Function Representation

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1. Representation for canonical “AND-OR” expression (Minterm)

When the logical function is a canonical "AND-OR" expression, simply find the small squares corresponding to the Minterm in the expression on the Karnaugh map and fill them with 1 and the remaining small squares are filled with 0 can be obtained. The Karnaugh map of the function.

E.g., a three-variable function $F(A, B, C) = \sum m(1, 2, 3, 7)$ its K map is represented as:

		AB			
		00	01	11	10
C	0	0	1	0	0
	1	1	1	1	0

$$F(A, B, C) = \sum m(1, 2, 3, 7)$$

2. Representation for common "AND-OR" expressions

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When the logic function is a common "AND-OR" expression, its Karnaugh map can be made based on the commonality of "AND" and the superposition of "OR".

E.g., a 4 variable function:

$$F(A,B,C,D) = AB + CD + \overline{A}\overline{B}C$$

its K map is:

		AB			
		00	01	11	10
CD	00	0	0	1	0
	01	0	0	1	0
	11	1	1	1	1
	10	1	0	1	0

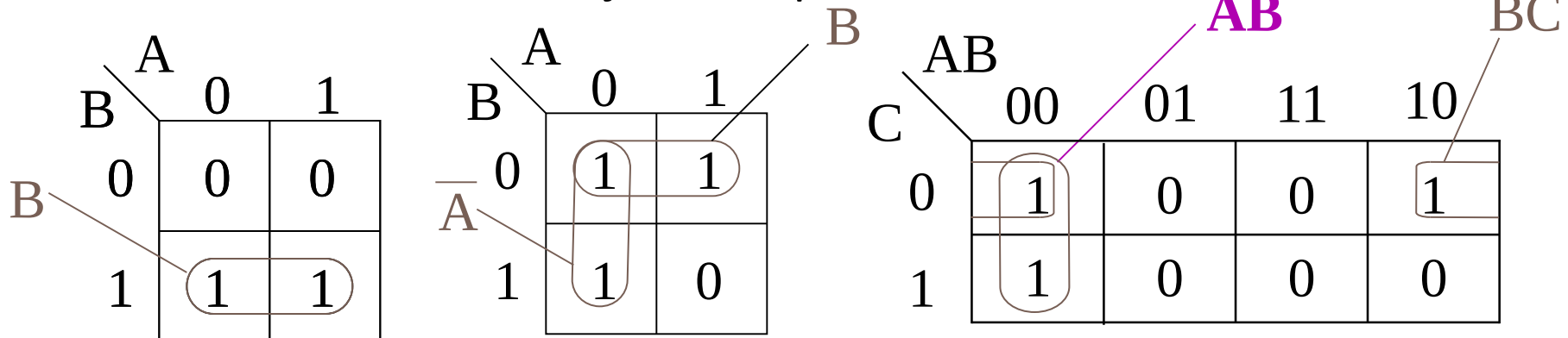
For the convenience, a square filled with 1 is usually called "1 squares". The square filled with 0 is called the 0 square. 0 squares are sometimes left blank.

IV. Combining Squares in K map

When a function is represented by a Karnaugh map, which of the smallest items can be combined? We use the Karnaugh maps of 2, 3, and 4 variables as examples as follows.

1. If two squares are adjacent, or at the end of a row (column), the two squares can be merged as a K rectangle, and a variable can be eliminated after the merger.

E.g., the following 2,3-variable K map shows the typical situations of the two adjacent squares.



2. Four adjacent squares can form a large square; in addition, if the four squares are at a row, or at a column, or at the end of two rows (columns), or at the four corners; two variables can be eliminated.

E.g, the following 3-variable K map shows the typical situations:

		AB			
		00	01	11	10
C	0	1	0	0	1
	1	1	0	0	1

Diagram illustrating a 3-variable Karnaugh map for variables A, B, and C. The map shows two rows (C=0 and C=1) and four columns (AB=00, 01, 11, 10). The values in the cells are 1, 0, 0, 1 for C=0 and 1, 0, 0, 1 for C=1. A pink line connects the two 1s in the first column (AB=00) and the two 1s in the last column (AB=10), indicating a group of four 1s. This group is labeled $\overline{B}$ in pink, indicating that B is eliminated.

		AB			
		00	01	11	10
C	0	0	1	1	0
	1	0	1	1	0

Diagram illustrating a 3-variable Karnaugh map for variables A, B, and C. The map shows two rows (C=0 and C=1) and four columns (AB=00, 01, 11, 10). The values in the cells are 0, 1, 1, 0 for C=0 and 0, 1, 1, 0 for C=1. A pink line connects the two 1s in the third column (AB=01) and the two 1s in the fourth column (AB=11), indicating a group of four 1s. This group is labeled B in pink, indicating that B is eliminated.

The 4-variable K map's typical situations

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$\overline{B}\overline{D}$ BD

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

$\overline{B}D$ $B\overline{D}$

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

$\overline{A}B$ CD

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	1	0	0
11	1	1	1	1
10	0	1	0	0

3. Eight squares form a large K rectangle, in two adjacent rows (columns), or in two side rows (columns), the Minterm expression can be merged, and the three variables can be eliminated after the merger.

E.g. the following 3-4 variable K maps show the two typical situations of eight adjacent squares.

		AB			
		00	01	11	10
C	0	1	1	1	1
	1	1	1	1	1

(a)

		AB			
		00	01	11	10
CD	00	1	1	1	1
	01	0	1	1	0
	11	0	1	1	0
	10	1	1	1	1

(b)

The summary of the K rectangle with n variables:

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(1) The amount number of K rectangle must be 2^m , m is a integer number that is less or equal to n

(2) In a K rectangle with 2^m squares, the Minterms in the rectangle have m different variables, and $(n-m)$ identical variables.

(3) Assume a K rectangle with 2^m squares, its corresponding logic algebra expression can be represented as an “AND” term with $(n-m)$ variables, the AND term is composed of the 2^m squares with the identical variables in the rectangle.

(4) if $m = n$, the K rectangle can cover the entire K map, the rectangle can be represented by 1, i.e., the fourth property in the Minterm.

IV, K Map Simplification

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1. Important concepts

Implicant: In the function's AND-OR expression, each AND item is called the function's Implicant.

In a function's K map, each square corresponding to a Minterm is a implicant.

Prime Implicant: a product term obtained by combining the maximum possible number of adjacent squares in the map into a rectangle with the number of squares a power of 2.

In a K map, if a rectangle cannot be covered by any other larger rectangles, the rectangle is defined as the prime implicant.

Essential Prime Implicant: A prime implicant is called an Essential Prime Implicant, if it is the only prime implicant that covers (includes) one or more minterms, which are not covered by any other prime implicant.

In a K map, if a K rectangle contains a square or more one square that is not shared by other rectangles, the K rectangle is called the EPI.

2. The general steps to find the simplest AND-OR expression for logical functions

Step 1: Draw the logic function's K map;



Step 2: Draw all the K rectangles in the K map



Step 3: Find the EPI from all the K rectangles



Step 4: Find the set of simplest EPI and minterms

If the K rectangles cannot cover all the squares with value 1, we will use the put the remainder single squares into the set.

E.g. Use the K map to simplify the following logic function $F(A,B,C,D) = \sum m(0,3,5,6,7,10,11,13,15)$

Solution:

AB \ CD		00	01	11	10
00	1				
01			1	1	
11	1	1	1	1	1
10			1		1

In the map, the five EPI covers all the squares with value 1; therefore, we can Oring each EPI's corresponding AND terms, then we can have the simplest “AND-OR” expression of F:

$$F(A,B,C,D) = \overline{A}\overline{B}\overline{C}\overline{D} + \overline{A}BC + \overline{A}\overline{B}C + BD + CD$$

E.g. Use the K map to simplify the following logic function

$$F(A, B, C, D) = \sum m(2, 3, 6, 7, 8, 10, 12)$$

Solution

AB \ CD		AB			
		00	01	11	10
CD	00			1	1
	01				
	11	1	1		
	10	1	1		1

i.e. $F(A, B, C, D) = \overline{A}C + A\overline{C}\overline{D} + A\overline{B}\overline{D}$

or $F(A, B, C, D) = \overline{A}C + A\overline{C}\overline{D} + \overline{B}C\overline{D}$

where, the two "AND-OR" function are of the same complexity.
Thus, the simplest AND-or expression of a function is not necessarily unique.

Use the K map to simplify the following logic function

$$F(A, B, C, D) = \sum m(1, 3, 5, 7, 8, 9, 10, 11, 14, 15)$$

Solution

		AB			
		00	01	11	10
CD	00	0	0	0	1
	01	1	1	0	1
	11	1	1	1	1
	10	0	0	1	1

$\bar{A}D$ (red line pointing to the 2x2 group of 1s in the first two columns)

$A\bar{B}$ (orange line pointing to the 2x1 group of 1s in the last column)

AC (yellow line pointing to the 2x2 group of 1s in the last two columns)

Thus, F's simplest “AND-OR” expression is

$$F(A, B, C, D) = \bar{A}D + A\bar{B} + AC$$

The principles of simplification of using the K map:

- (1) The number of K rectangle should be minimized on the precondition of covering all the smallest terms in the function;**
- (2) The K rectangles should reach its maximum under the premise of satisfying the merger laws;**
- (3) According to the needs of the merge, each Mintem can be surrounded by multiple K rectangles**

3. The general steps for finding the simplest “OR-AND” expression of a logical function

● When the given logical function is an AND-OR expression or a canonical AND-OR expression, the “double-complement method” is usually used.

Steps:

(1) Make a K map of F , find the simplest AND-OR expression of the complement function (merge the 0 square on the K map);

(2) Complement the obtained simplest AND-OR expression, to yield the simplest OR-AND expression of function F .

E.g., use the K map to solve the following function's

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10)$$

Simplest “AND-OR” terms

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Solution: obtain the simplest AND-OR expression first.

$$\overline{F} = AB + CD + B\overline{D}$$

		AB			
		00	01	11	10
CD	00	1	0	0	1
	01	1	1	0	1
	11	0	0	0	0
	10	1	0	0	1

Then complement the both sides of the function $\overline{F} = AB + CD + B\overline{D}$

Then, we can obtain the simplest OR-AND expression for a function

$$F = \overline{\overline{F}} = (\overline{A} + \overline{B}) \cdot (\overline{C} + \overline{D}) \cdot (\overline{B} + D)$$

● When the given logical function is an “OR-AND” expression or a canonical “OR-AND” expression, the “double dual method” is usually used.

Steps:

(1) Make the K map of F' and find the simplest AND-OR expression for F' ;

(2) Do another dual process to F' (the simplest “AND-OR” expressions), then the simplest “OR-AND” expression of the function F is obtained.

The Karnaugh Map that can simplify logic functions has the advantages of convenience, intuition, and easy to learn. However, due to the constraints of the variable number, when the number of variables is greater than 6, the drawing and the recognition of the graphics become extremely complicated.

In order to overcome the shortcomings of Karnaugh map, another simplification method – Quine-McCluskey tabular method was introduced.

(Read textbooks for more details on Quine-McCluskey tabular)